



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16240
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2785.66
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10726997	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16240
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

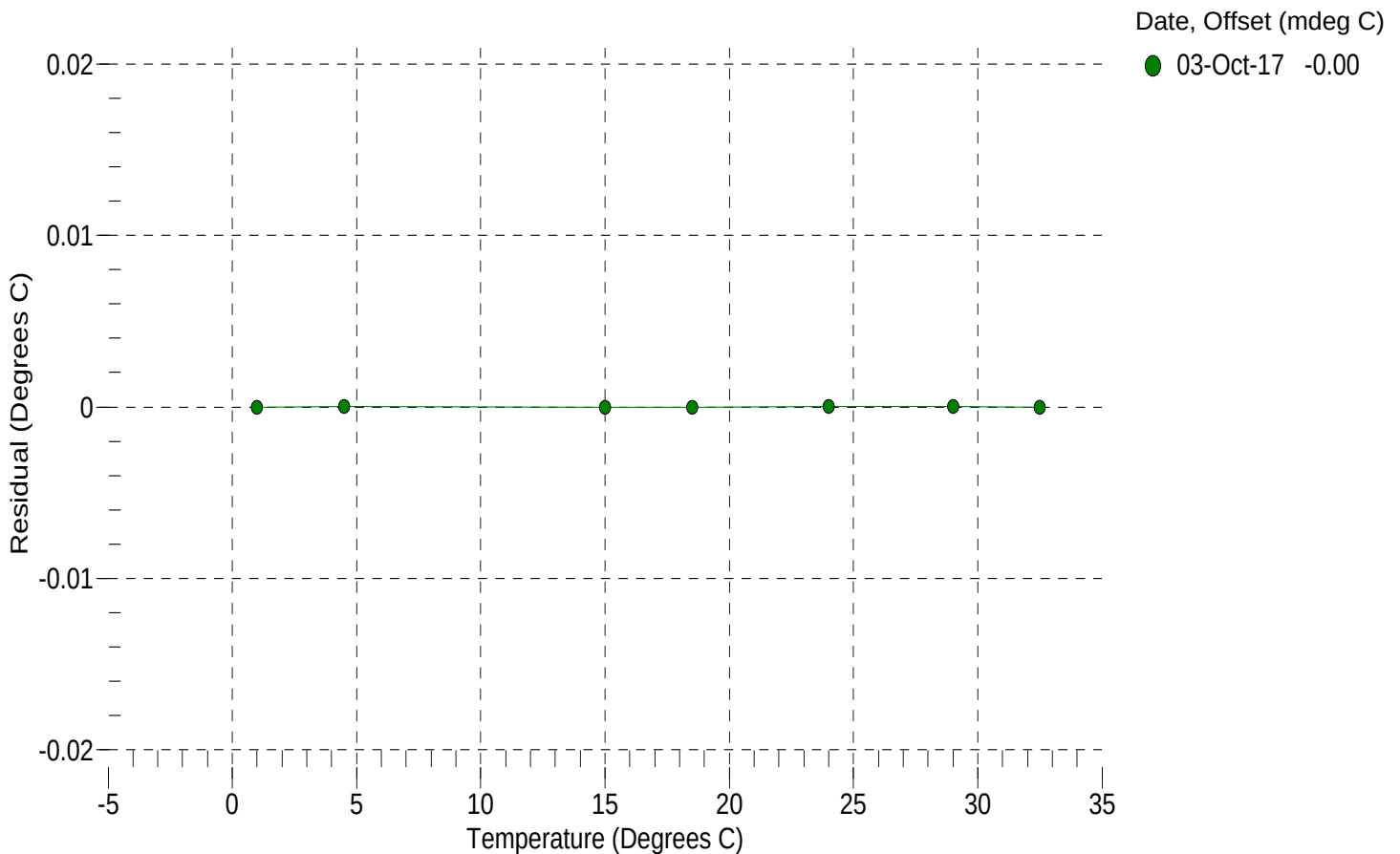
a0 = -2.536825e-004
a1 = 3.267064e-004
a2 = -5.460327e-006
a3 = 2.291108e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	564575.1	1.0000	-0.0000
4.5000	484858.0	4.5000	0.0000
15.0000	313073.6	15.0000	-0.0000
18.5000	272255.4	18.5000	-0.0000
24.0000	219878.4	24.0000	0.0000
29.0000	182148.1	29.0000	0.0000
32.5000	160182.1	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.966721e-001
h = 1.283129e-001
i = 1.615172e-006
j = 1.555294e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 3.8886e-007

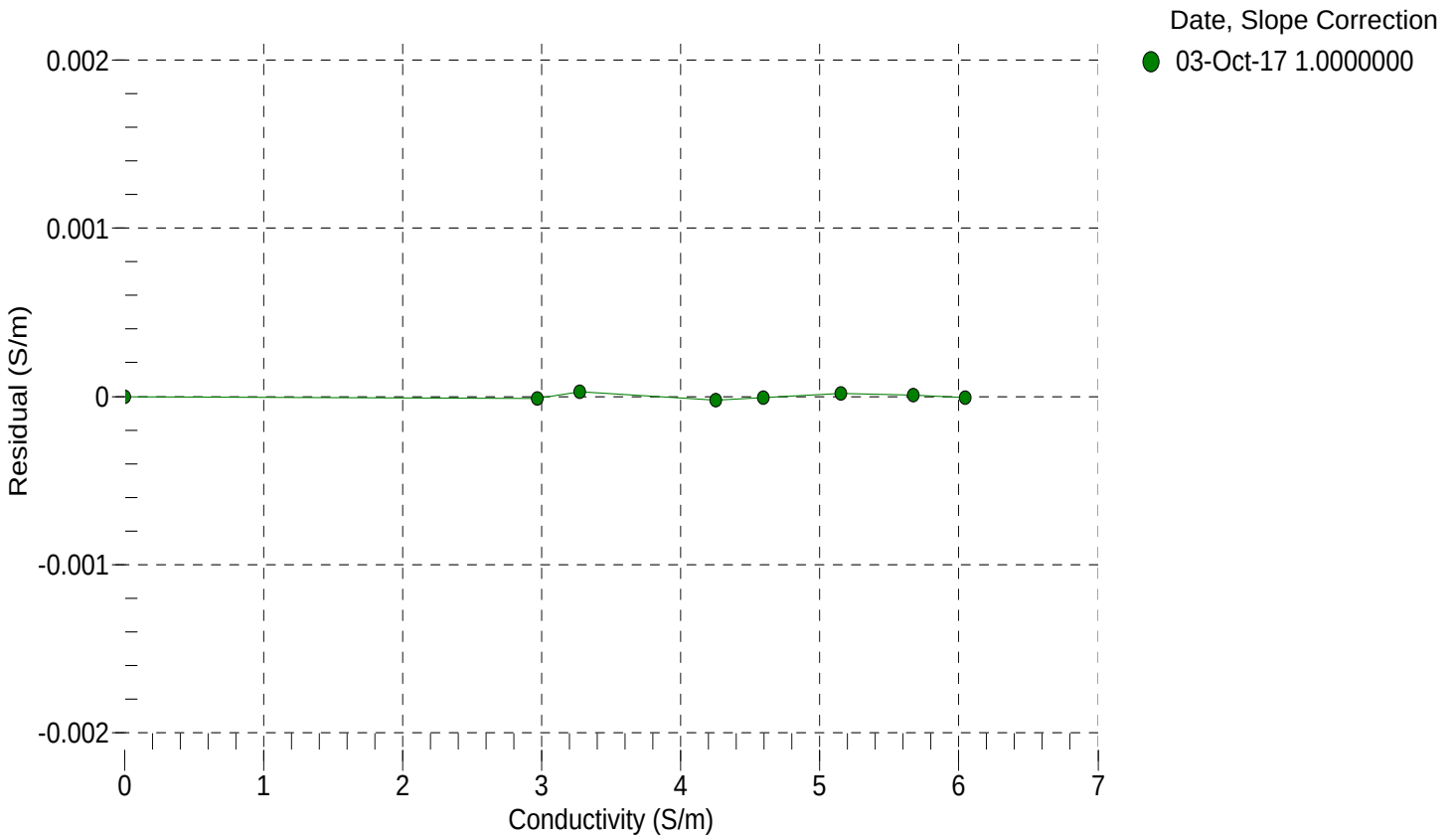
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2785.66	0.00000	0.00000
1.0000	34.6932	2.96644	5547.02	2.96643	-0.00001
4.5000	34.6732	3.27255	5756.47	3.27258	0.00003
15.0000	34.6306	4.25124	6379.38	4.25121	-0.00002
18.5000	34.6218	4.59534	6584.18	4.59534	-0.00001
24.0000	34.6123	5.15162	6902.18	5.15164	0.00002
29.0000	34.6073	5.67193	7186.59	5.67193	0.00001
32.5000	34.6050	6.04330	7382.73	6.04329	-0.00001

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

$\text{Conductivity (S/m)} = (g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16240
CALIBRATION DATE: 27-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10726997

COEFFICIENTS:

PA0 =	1.513912e-001	PTCA0 =	5.237531e+005
PA1 =	9.129216e-003	PTCA1 =	8.690422e-001
PA2 =	-1.506643e-010	PTCA2 =	-7.765812e-005
PTEMPA0 =	-5.848018e+001	PTCB0 =	3.062345e+005
PTEMPA1 =	5.360125e-002	PTCB1 =	1.177837e+002
PTEMPA2 =	-4.713434e-007	PTCB2 =	-2.302220e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.63	525373.6	1541.3	14.69	0.00	32.50	1723	525462.89
591.69	588941.0	1544.7	591.56	-0.00	29.00	1656	525462.39
1168.83	652685.2	1545.4	1168.82	-0.00	24.00	1560	525459.22
1746.03	716560.4	1546.0	1746.06	0.00	18.50	1455	525452.32
2323.18	780561.2	1546.9	2323.20	0.00	15.00	1388	525446.85
2900.19	844672.1	1547.4	2900.11	-0.00	4.50	1188	525440.71
2323.21	780569.9	1546.9	2323.28	0.00	1.00	1121	525437.22
1746.04	716564.6	1546.5	1746.09	0.00	TEMPERATURE (°C) SPAN (mV)		
1168.98	652705.7	1545.9	1169.01	0.00			
591.65	588939.0	1545.8	591.54	-0.00			
14.62	525373.9	1548.0	14.69	0.00			
					5.50	306812.44	
					25.70	307740.91	
					34.05	307575.87	

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

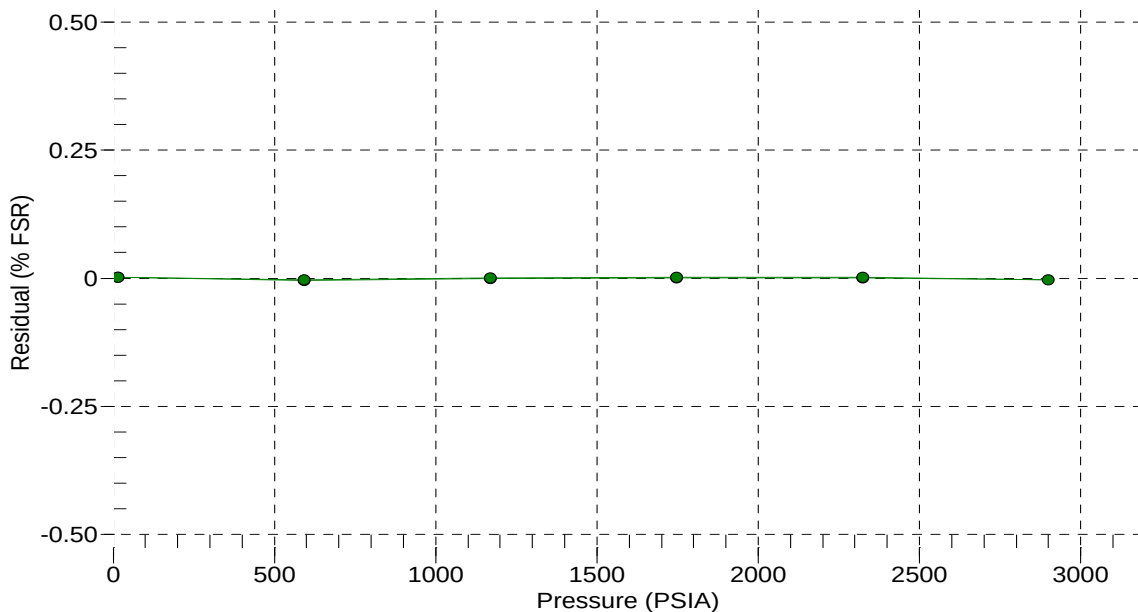
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 27-Sep-17 -0.00





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Pressure Test Certificate

Test Date: **2017-10-09**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16240**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **TH**

